



Department of Mechanical Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. Mechanical Engineering

Course Code & Title: ME8594 Dynamics of Machines

Name of the Faculty member (s): Dr.J.Jabinth, AP/Mechanical

Innovative Practice Description

- **Unit / Topic: Unit II / Dynamic Analysis in reciprocating engines using MATLAB**
- **Course Outcome: CO2**
- **Date: 21.10.2021**
- **Topic Learning Outcome: TLO8**
- **Activity Chosen: Interactive learning using MATLAB**
- **Justification:** Students find the velocity, angular velocity, acceleration and angular acceleration of a complex problem.
- **Time Allotted for the Activity: 30 minutes**
- **Details of the Implementation:** A MATLAB code is written to plot the angular velocity and angular acceleration of the slider crank mechanism. This code is demonstrated to student's line by line and explained them the advantage of using MATLAB to determine the Angular velocity and Angular acceleration at every instant of degree.
- **CO – PO / PSO mapping:**

PO	PO1	PO2	PO12	PSO2
CO5	3	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO12	PSO4
	3	2	3	2
Justification for correlation	Students must have fundamental knowledge in the topics learnt in subject	Students will be able to understand the unbalancing problem, SDOF and its consequence of vibration.	Students can learn more like balancing of reciprocating engines, Multi Degree of Freedom etc.	Provides manufacturing solution while manufacturing a component.



- **Program code (MATLAB):**

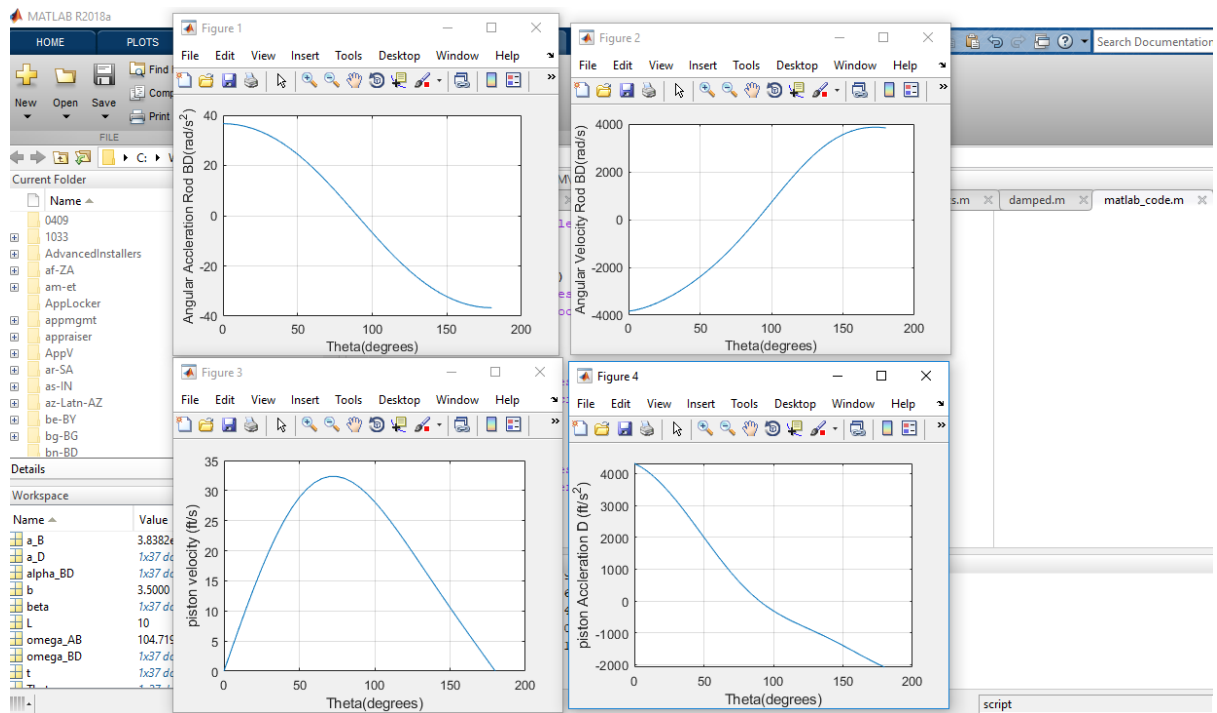
```
L=10;
b=3.5;
omega_AB=1000*2*pi/60;
v_B=b*omega_AB;
Theta=[0:5:180];
t=Theta*pi/180;
beta=asin((b/L)*sin(t));
v_D=v_B*sin(t+beta)./cos(beta);
v_D=v_D/12;
omega_BD=v_B*cos(t)./(L*cos(beta));
a_B=b*omega_AB^2;
alpha_BD=(L.*omega_BD.^2.*sin(beta)-a_B.*cos(t))./(L.*cos(beta));
a_D=a_B.*cos(t)+L.*omega_BD.^2.*cos(beta)+L.*alpha_BD.*sin(beta);
a_D=a_D/12;
z=[Theta;omega_BD;alpha_BD;v_D;a_D];
fprintf('Theta Angular Angular Piston Piston\n')
fprintf('velocity Accleration velocity acceleration\n')
fprintf('(deg) (rad/s) (rad/s^2) (ft/s) (ft/s^2)\n')
fprintf('\n');
fprintf('%5.3f %6.3f %6.3f %6.3f %6.3f\n',z);
fprintf('\n');
%fprintf('The two values of theta for zero collar speed are %5.3f and %5.3f degrees\n',a1,a2)
figure(1)
plot(Theta,omega_BD)
xlabel('Theta(degrees)')
ylabel('Angular Accleration Rod BD(rad/s^2)')
grid on
figure(2)
plot(Theta,alpha_BD)
xlabel('Theta(degrees)')
ylabel('Angular Velocity Rod BD(rad/s)')
grid on
figure(3)
plot(Theta,v_D)
xlabel('Theta(degrees)')
ylabel('piston velocity (ft/s)')
grid on
figure(4)
plot(Theta,a_D)
xlabel('Theta(degrees)')
ylabel('piston Accleration D (ft/s^2)')
grid on
```



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OUTPUT



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Based on the feedback received from students, they told this method of teaching is very interesting.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Students can be able to solve any kind of complex problems related to reciprocating components.

- ❖ **Challenges faced in implementation:**

- MATLAB installation in system.
 - Writing program in matlab.

References:

- ❖ https://in.mathworks.com/?s_tid=gn_logo
- ❖ Khurmi, R.S., "Theory of Machines" 14th edition, S chand publications 2005.
- ❖ Rattan, S.S, "Theory of Machines", 4th Edition, Tata McGraw-Hill, 2014.

Signature of Faculty Member

HOD